Bank Credit Limits, Alternative Finance, and Firm Investment

An access-cost threshold model

Final presentation — work in progress

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github.com/amchavezu/ai-project

A firm has a profitable project, but its bank will lend only up to a limit. After exhausting that line, the firm faces a choice:

stop investing or pay more for another source of funds?

The answer depends on one comparison:

marginal return at the bank limit versus cost of alternative finance.

Access is not simply "yes" or "no": its cost determines the choice.

Question. How does the cost of accessing alternative finance affect investment and output for a firm facing a bank-credit limit?

Answer. For a firm that would exhaust its bank line at the bank financing cost, there is an access-cost threshold:

- above it, the firm stops at the bank limit, so credit availability determines investment;
- below it, the firm invests beyond the limit, so the marginal financing cost determines investment.

The model derives the threshold, the two optimal choices, and their comparative statics.

The thesis finds a preliminary qualitative pattern: credit volume and a credit spread may contain different forecasting information across Peruvian sectors.

What this paper does

It derives a mechanism linking financing access to investment responses and states cross-sectional predictions that the thesis can test.

What this paper does not do

It does not derive forecast accuracy, causal effects, or the horizon at which an indicator should peak.

At date t, a firm with internal funds n committed to the project chooses capital k, bank credit b, and alternative finance m. Capital produces at $t+1$.

$$\max_{k,b,m} \ \beta F(k) - R_B b - (R_B + \tau)m \quad \text{s.t.} \quad k = n + b + m, \quad 0 \le b \le \bar{b}, \quad m \ge 0.$$

$$F' > 0, \quad F'' < 0, \quad \tau \ge 0.$$

Bank finance is cheaper but capped. R_B and $R_B + \tau$ are present-value costs per financed unit. The per-unit wedge τ captures issuance, information, and risk-premium costs outside the bank.

Let $\bar{k} = n + \bar{b}$ be capital after the bank line is exhausted. Focus on a firm that still wants to invest at that point at the bank rate:

$$\beta F'(\bar{k}) > R_B.$$

The next unit costs $R_B + \tau$ outside the bank. It is profitable when

$$\beta F'(\bar{k}) > R_B + \tau.$$

Therefore the threshold is

$$\boxed{\tau^* = \beta F'(\bar{k}) - R_B > 0.}$$

This single threshold replaces the earlier binary access/no-access cases.

	Expensive access	Affordable access
Condition	$\tau \geq \tau^*$	$0 \leq \tau < \tau^*$
Capital	$k^* = \bar{k} = n + \bar{b}$	$\beta F'(k^*) = R_B + \tau$
Bank-line effect	$\partial k^*/\partial \bar{b} = 1$	$\partial k^*/\partial \bar{b} = 0$
Access-cost effect	$\partial k^*/\partial \tau = 0$ locally	$\partial k^*/\partial \tau = 1/[\beta F''(k^*)] < 0$
What determines scale?	bank-credit limit	marginal financing cost

At $\tau = \tau^*$ the policy is continuous but has a kink. Comparative statics are local to a strict side of the threshold.

7 · Why the proposition is true

After substituting the financing identity, profit has two slopes:

$$\Pi'(k) = \begin{cases} \beta F'(k) - R_B, & k < \bar{k}, \\ \beta F'(k) - (R_B + \tau), & k > \bar{k}. \end{cases}$$

At the bank limit, the right-hand slope is

$$\Pi'_+(\bar{k}) = \beta F'(\bar{k}) - R_B - \tau = \tau^* - \tau.$$

- If $\tau \geq \tau^*$, the slope turns non-positive at the kink.
- If $\tau < \tau^*$, profit keeps increasing beyond the bank limit.
- Strict concavity makes the resulting optimum global and unique.

Hard step: proving that the kink is the unique global optimum, not only a local sign change; strict concavity completes that step.
Status: the firm-level result and the sectoral formula are derived. The empirical distributions G_s remain to be estimated.

Let

$$F(k) = ak - \frac{c}{2}k^2, \qquad a, c > 0.$$

Then

$$\tau^* = \beta\big[a - c(n + \bar{b})\big] - R_B,$$

and the outside-finance target is

$$k_M(\tau) = \frac{\beta a - R_B - \tau}{\beta c}.$$

Within the focal domain,

$$\boxed{k^*(\tau) = \max\{n + \bar{b},\ k_M(\tau)\}}.$$

Equivalently,

$$\boxed{k^*(\tau) - \bar{k} = \max\left\{0, \frac{\tau^* - \tau}{\beta c}\right\}}.$$

The distance below the threshold maps directly into additional investment and, because $F' > 0$, additional output.

For $\beta = .95$, $a = 2$, $c = .5$, $n = .4$, $\bar{b} = .5$, and $R_B = .5$:

$$\bar{k} = .900, \qquad \tau^* = .9725.$$

	Expensive access	Affordable access
Access wedge	$\tau = 1.10$	$\tau = .90$
Optimal capital	$k^* = .900$	$k^* = 1.053$
Output	$F(k^*) = 1.598$	$F(k^*) = 1.828$
Larger bank line	raises capital locally	changes funding mix locally
Lower τ	no local scale effect	raises capital and output

At $\tau = .90$, affordable access adds 0.153 units of capital and 0.231 units of output relative to stopping at the bank limit.

The interesting object is the change in behavior at τ^*, not the chosen calibration.

Let firms in sector s draw access wedges from $G_s(\tau)$. The constrained share is generated by the threshold:

$$\lambda_s = \Pr(\tau_i \geq \tau^*) = 1 - G_s(\tau^*).$$

- A distribution shifted toward high wedges produces more firms whose investment is governed by the bank limit.
- A distribution shifted toward low wedges produces more firms whose investment is governed by marginal financing costs.
- Tradable and non-tradable labels enter only here, as an empirical hypothesis about G_T and G_N.

The model no longer assumes directly that an entire sector is one firm type.

The mechanism suggests sharper tests than comparing forecast peaks:

1. Credit-quantity effects should be stronger among firms with exhausted bank lines and high alternative-finance wedges.
2. Financing-cost effects should be stronger among firms already using bonds, foreign credit, or other non-bank sources.
3. A larger bank line should raise total finance above τ^*, but should mainly change debt composition below τ^*.

The thesis's credit volume and spread are proxies for $\bar{b}$ and the relevant financing wedge. The mapping must be tested rather than assumed.

`code/verify.py` verifies:

- the formula for τ^* and the identity $k_M(\tau^*) = \bar{k}$;
- the first-order condition on each side of the threshold;
- continuity and the change in local derivatives at the kink;
- every numerical value reported in the paper and slides.

It cannot establish:

- that G_N is shifted above G_T in Peru;
- that the observed spread equals τ;
- causality or exact forecast horizons.

Paper statement:

$$\tau < \tau^* \quad \Longleftrightarrow \quad \bar{k} < k_M(\tau) \qquad \text{(quadratic technology)}.$$

Corresponding Lean declaration:

```
theorem access_below_threshold_iff
  (hbeta :  0 < beta) (hc :  0 < c) :
  tau < accessThreshold beta a c RB cap <->
    cap < targetCapital beta a c (RB + tau)
```

The prototype states the algebraic equivalence under the same positivity assumptions used in the quadratic model; it adds no assumption containing the conclusion. A clean end-to-end AppliedModelingLib check is still pending. Lean cannot test the empirical interpretation of τ or the sectoral distribution.

```
have hden :  0 < beta * c := mul_pos hbeta hc
unfold accessThreshold targetCapital
constructor
  .  intro h
    apply (lt_div_iff₀ hden).2
    nlinarith
```

- Positivity permits multiplying by βc without reversing the inequality.
- Unfolding replaces the economic names by their algebraic definitions.
- In this proof, `nlinarith` discharges the equivalence between the threshold inequality and the capital-target inequality.

The reverse implication repeats the same steps in the opposite direction.

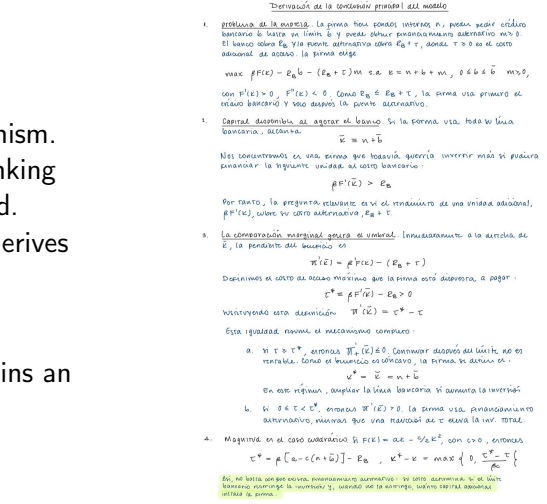
Initial AI claim:

$$\lambda_N > \lambda_T$$

was treated as the sectoral mechanism.
Problem found by hand: the ranking was assumed rather than generated.
Verdict: rejected. The revision derives

$$\lambda_s = 1 - G_s(\tau^*).$$

Whether G_N differs from G_T remains an empirical hypothesis.



Derivación de la concesión primera del modelo

1. problema de la empresa. La firma tiene pocos intereses r_i, puede pedir crédito bancario b hasta un límite $\bar{b}$ y puede obtener financiamiento alternativo más $\tau \geq 0$. El banco cobra R_B y la fuente alternativa cobra $R_B + \tau$, donde $\tau \geq 0$ es el costo adicional de acceso. La firma elige

$$\max_{k,b,m} \; \phi F(k) - R_B b - (R_B + \tau) m \quad \text{s.a.} \quad k = n + b + m, \; 0 \leq b \leq \bar{b}, \; m \geq 0,$$

con $F'(k) > 0$, $F''(k) < 0$. Como $R_B < R_B + \tau$, la firma usa primero el crédito bancario y solo después la fuente alternativa.

2. Capital disponible al agotar el banco. Si la firma usa toda su línea bancaria, alcanza

$$\bar{k} \equiv n + \bar{b}.$$

Nos concentramos en una firma que todavía querría invertir más si pudiera financiar la siguiente unidad al costo bancario:

$$\phi F'(\bar{k}) > R_B.$$

Por tanto, la pregunta relevante es si el financiamiento de una unidad adicional, $\phi F'(\bar{k})$, vale su costo alternativo, $R_B + \tau$.

3. La comparación marginal genera el umbral. Inmediatamente a la derecha de $\bar{k}$, la pendiente del beneficio es

$$\pi'(\bar{k}) = \phi F'(\bar{k}) - (R_B + \tau).$$

Definimos el costo de acceso máximo que la firma está dispuesta a pagar:

$$\tau^* \equiv \phi F'(\bar{k}) - R_B > 0$$

Reinterpretando esta definición: $\pi'(\bar{k}) = \tau^* - \tau$

Esta igualdad resume el mecanismo completo:

a. Si $\tau > \tau^*$, entonces $\pi'_+(\bar{k}) \leq 0$. Continuar después del límite no es rentable. Como el beneficio es cóncavo, la firma se detiene en

$$k^* = \bar{k} = n + \bar{b}$$

En este régimen, ampliar la línea bancaria sí aumenta la inversión.

b. Si $0 \leq \tau < \tau^*$, entonces $\pi'(\bar{k}) > 0$. La firma usa financiamiento alternativo, mientras que una reducción de τ eleva la inv. total.

4. Magnitud en el caso cuadrático. Si $F(k) = ak - \frac{c}{2}k^2$, con $c > 0$, entonces

$$\tau^* = \phi\left[a - c(n + \bar{b})\right] - R_B, \quad k^* - \bar{k} = \max\left\{0, \frac{\tau^* - \tau}{\phi c}\right\}$$

Así, no basta con que exista financiamiento alternativo: su costo determina si el límite bancario restringe la inversión y, cuando lo restringe, cuánto capital adicional utilizaría la firma.

Known limits

- $\bar{b}$ and τ remain reduced-form primitives;
- the observed loan spread may not equal the alternative-finance wedge;
- the static model does not generate forecast horizons or causal effects.

Questions for the class

1. Is the observed spread a defensible proxy for τ, or should the empirical mapping use another financing-cost measure?
2. Which observable best disciplines G_s: firm size, exporter status, bond issuance, or foreign-credit access?
3. Should the next extension endogenize $\bar{b}$ through net worth, or add investment dynamics first?